Experiment Number: 1

Simulation and Comparative Analysis of FDMA, TDMA and CDMA: Resource Allocation, User Capacity and Mobile-Generation Context

Aim:To simulate and compare the basic resource-allocation methods used in FDMA, TDMA and CDMA, and examine how their assumed user capacity and interference behaviour differ.

Objective 3: Mobile-generation comparison
To relate representative multiple-access techniques to the evolution of mobile communication systems from 1G to 5G.

CODE:

```
% Experiment 1: Mobile-Generation Comparison
% Program 3

clear;
clc;
close all;

% Display heading
disp('MOBILE COMMUNICATION GENERATIONS');
disp(' ');

% Display table headings
fprintf('%-12s %-12s %-18s %-35s\n', ...
    'Generation', 'System', 'Access Method', 'Key Idea');

fprintf('%s\n', repmat('-', 1, 82));

% Generation 1
fprintf('%-12s %-12s %-18s %-35s\n', ...
    '1G', 'AMPS', 'FDMA', 'Separate frequency channels');

% Generation 2 - GSM
fprintf('%-12s %-12s %-18s %-35s\n', ...
    '2G', 'GSM', 'TDMA', 'Separate time slots');

% Generation 2 - IS-95
fprintf('%-12s %-12s %-18s %-35s\n', ...
```

```
    '2G', 'IS-95', 'CDMA', 'Code-based user separation');

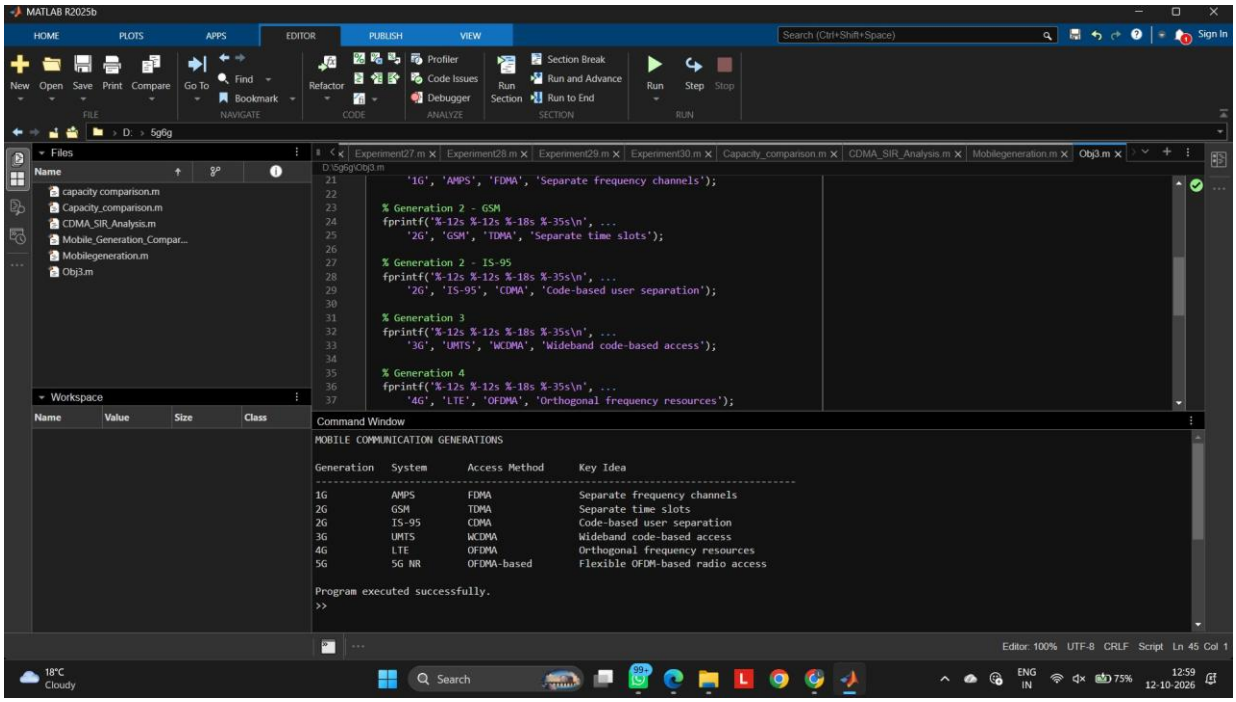
% Generation 3
fprintf('%-12s %-12s %-18s %-35s\n', ...
    '3G', 'UMTS', 'WCDMA', 'Wideband code-based access');

% Generation 4
fprintf('%-12s %-12s %-18s %-35s\n', ...
    '4G', 'LTE', 'OFDMA', 'Orthogonal frequency resources');

% Generation 5
fprintf('%-12s %-12s %-18s %-35s\n', ...
    '5G', '5G NR', 'OFDMA-based', 'Flexible OFDM-based radio access');

disp(' ');
disp('Program executed successfully.');
```

OUTPUT:



Performance Evaluation Rubric

Assessment Criterion	4 — Strong	3 — Meets	2 — Partial	1 — Limited
Objective 3: Mobile-generation comparison	Correctly links generations, systems and access methods; distinctions explained.	Mostly accurate, with minor omissions or imprecise descriptions.	Several entries incomplete or unclear.	Table missing or contains major inaccuracies.
Parameter study and recorded results	Changes relevant parameters, reruns applicable programs, records outputs and explains effects.	Provides rerun results and comparison, but one change or explanation is incomplete.	Makes limited changes or records results without clear comparison.	No meaningful parameter study or usable results record.
Analysis and interpretation	Answers accurately, uses results as evidence and distinguishes model behavior from real-system behavior.	Most answers sound and refer to results, with some gaps in explanation.	Answers partly correct but mainly describe outputs without explaining them.	Analysis missing, unsupported or substantially incorrect.
Engineering judgment and conclusion	Gives a clear conclusion, identifies assumptions and limitations, and avoids unsupported performance rankings.	Conclusion is reasonable but omits a relevant assumption, limitation or implication.	Conclusion mostly repeats results, with little discussion of limitations.	Conclusion missing or draws unsupported conclusions.
Report organization and evidence	Includes code, tables, required figures, parameter values and clear captions.	Required evidence present, with minor organization or labeling issues.	Some evidence missing or difficult to follow.	Report substantially incomplete or unclear.
Individual explanation / viva	Clearly explains own work, formulas and results.	Explains most aspects, with minor gaps.	Gives partial explanations but struggles to connect code and results.	Cannot explain key parts of the submitted work.